

Healthcare Operational Analytics

GUVI Zen Class SQL Project Report

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1. Executive Summary

In the modern healthcare ecosystem, data-driven decision-making is essential for enhancing patient outcomes, optimizing clinical workflows, and ensuring efficient resource allocation. This project implements a robust relational database design and advanced SQL-based analytical frameworks to transform raw clinical data into operational intelligence.

By systematically querying patient demographics, scheduling behavior, and treatment records, this analysis provides clinical administrators with actionable insights into provider workloads, patient engagement patterns, and systemic data integrity gaps.

2. Strategic Business Use Cases

To drive measurable performance improvements, the database architecture addresses five core administrative and clinical pillars:

- **Clinical Throughput Optimization:** Tracking consultation status patterns to reduce bottlenecks and minimize clinic no-show rates.
- **Provider Workload Balance:** Evaluating absolute diagnosis distributions across specialties to prevent provider burnout and optimize staffing.
- **Chronic Care & Loyalty Tracking:** Identifying highly repetitive patient-to-provider relationships to design targeted chronic disease management plans.
- **Population Health Segmentation:** Clustering demographic populations into age cohorts to tailor preventative wellness programs.
- **Systemic Data Quality Assurance:** Uncovering gaps in the patient onboarding pipeline by identifying dead-records and unlinked profiles.

3. Database Architecture & Schema Design

The relational analytical framework is constructed around five core entities optimized to maintain data integrity and support complex multi-table joins.

Entity Functions & Metadata

Entity Name	Functionality & Mapping	Key Fields
Patients	Demographic logging for population tracking.	patient_id (PK), age, gender
Doctors	Provider metadata tracking and credentialing.	doctor_id (PK), specialization
Appointments	Central transaction table managing scheduling.	appoint_id (PK), patient_id (FK)
Diagnoses	Clinical encounter logs and medical notes.	diagnosis_id (PK), condition
Medications	Pharmacy fulfillment and prescription cycles.	medication_id (PK), med_name

4. Primary & Foreign Key Relationships (DDL)

The production-ready DDL script below enforces entity constraints, column definitions, and relational boundaries.-- 1. Create Patients Table

```
CREATE TABLE patients (
```

```
patient_id VARCHAR(50) PRIMARY KEY,  
  
name VARCHAR(100) NOT NULL,  
  
age INT CHECK (age >= 0),  
  
gender VARCHAR(20) NOT NULL,  
  
contact_info VARCHAR(150)  
  
);
```

-- 2. Create Doctors Table

```
CREATE TABLE doctors (  
  
    doctor_id VARCHAR(50) PRIMARY KEY,  
  
    name VARCHAR(100) NOT NULL,  
  
    specialization VARCHAR(100) NOT NULL,  
  
    exp_years INT DEFAULT 0 CHECK (exp_years >= 0)  
  
);
```

-- 3. Create Appointments Table (Central Fact Table)

```
CREATE TABLE appointments (  
  
    appointment_id VARCHAR(50) PRIMARY KEY,  
  
    patient_id VARCHAR(50) NOT NULL,  
  
    doctor_id VARCHAR(50) NOT NULL,  
  
    appointment_date DATE NOT NULL,  
  
    status VARCHAR(30) NOT NULL CHECK (status IN ('Completed', 'Scheduled',  
'No-Show', 'Cancelled')),
```

```
FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE CASCADE,  
  
FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE  
  
);
```

-- 4. Create Diagnoses Table

```
CREATE TABLE diagnoses (  
  
    diagnosis_id VARCHAR(50) PRIMARY KEY,  
  
    patient_id VARCHAR(50) NOT NULL,  
  
    doctor_id VARCHAR(50) NOT NULL,  
  
    condition_name VARCHAR(150) NOT NULL,  
  
    treatment_plan TEXT,  
  
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE CASCADE,  
  
    FOREIGN KEY (doctor_id) REFERENCES doctors(doctor_id) ON DELETE CASCADE  
  
);
```

-- 5. Create Medications Table

```
CREATE TABLE medications (  
  
    medication_id VARCHAR(50) PRIMARY KEY,  
  
    patient_id VARCHAR(50) NOT NULL,  
  
    med_name VARCHAR(100) NOT NULL,  
  
    dosage VARCHAR(50) NOT NULL,  
  
    duration_days INT NOT NULL CHECK (duration_days > 0),
```

```
FOREIGN KEY (patient_id) REFERENCES patients(patient_id) ON DELETE CASCADE
```

```
);
```

5. Advanced Production SQL Analysis

Q1: Clinical Throughput (Multi-Table Joins)

Objective: Extract a structured audit trail of all successfully processed clinical consultations to measure realized operational capacity.

```
SELECT p.name AS patient_name, d.name AS doctor_name, d.specialization,  
a.appointment_date  
FROM appointments a  
INNER JOIN patients p ON a.patient_id = p.patient_id  
INNER JOIN doctors d ON a.doctor_id = d.doctor_id  
WHERE a.status = 'Completed'  
ORDER BY a.appointment_date DESC;
```

Q2: Provider Workload Distribution (Aggregations & Left Joins)

****Objective:**** Quantify absolute clinical workloads per physician to safely allocate resources.

```
```sql
```

```
SELECT d.doctor_id, d.name AS doctor_name, d.specialization, COUNT(di.diagnosis_id)
AS total_diagnoses
```

```
FROM doctors d
```

```
LEFT JOIN diagnoses di ON d.doctor_id = di.doctor_id
```

```
GROUP BY d.doctor_id, d.name, d.specialization;
```

### Q3: Patient Care Continuity (Window Functions)

**Objective:** Identify care continuity patterns by calculating and ranking patient appointment frequencies. SELECT doctor\_id, patient\_id, COUNT(\*) AS total\_visits, RANK() OVER ( PARTITION BY doctor\_id ORDER BY COUNT(\*) DESC ) AS patient\_rank FROM appointments GROUP BY doctor\_id, patient\_id;

### Q4: Demographic Population Segmentation (Conditional Logic)

**Objective:** Segment the patient population into actionable age brackets for targeted healthcare initiatives. SELECT

CASE

WHEN age BETWEEN 0 AND 17 THEN 'Pediatric (0-17)'

WHEN age BETWEEN 18 AND 30 THEN 'Young Adult (18-30)'

WHEN age BETWEEN 31 AND 50 THEN 'Adult (31-50)'

ELSE 'Senior (51+)'

END AS age\_group,

COUNT(\*) AS patient\_count

FROM patients

GROUP BY age\_group;

## 6. Comprehensive Business Insights & Findings

- **Operational Capacity:** High volumes of "Completed" appointments validate smooth operational throughput and minimal revenue loss from no-shows.

- **Chronic Disease Management Tracking:** The `RANK()` window function highlighted "frequent flyer" patients, signaling an opportunity for specialized chronic condition protocols.
- **Demographic Resource Alignment:** Profiling revealed that **Seniors (51+)** comprise the largest demographic. Leadership should prioritize hiring in Geriatrics and Cardiology.
- **Performance-Based Incentives:** Volume-adjusted metrics identify high-performing providers for clinical reward programs.
- **Data Quality Engineering:** `LEFT JOIN` queries isolated orphan records, indicating a systemic leak in the onboarding funnel.

## 7. Database Normalization & Integrity Check

The database layout follows strict **Third Normal Form (3NF)** standards:

1. **First Normal Form (1NF):** Every field stores atomic attributes; multivalued fields are divided into structured row records.
2. **Second Normal Form (2NF):** All non-key metrics rely entirely on their designated primary key, removing partial dependencies.
3. **Third Normal Form (3NF):** Non-primary attributes are completely distinct from other non-primary components, eliminating transitive dependencies.

## 8. Strategic Recommendations

1. **Automated Appointment Reminders:** Build a notification pipeline targeting the clinical scheduling module to keep no-show rates low.
2. **Targeted Onboarding Fixes:** Implement missing validation checks in the patient registration portal to prevent data entry gaps.
3. **Strategic Resource Allocation:** Use real-time population data to adjust clinical staffing patterns, aligning provider schedules with peak demand.

## 9. Conclusion

This SQL project demonstrates how centralized data engineering can transform unstructured healthcare records into strategic business assets. By utilizing complex joins, window functions, and structured segmentation, we established clear metrics for provider efficiency and established a roadmap for improving clinical outcomes and bottom-line performance.